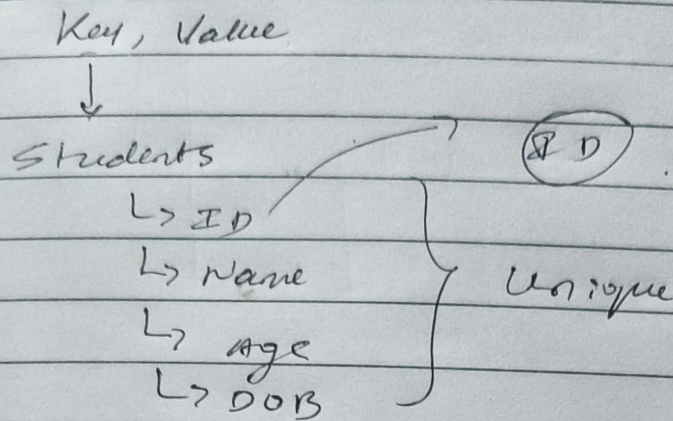


13 Dictionary

29/8/25

Day 42

Date :
Page No. :



Value :->

-> objects.

Set	Dict
value	key, values

Hashing -> (key)

→ Add
→ Del
→ Search

} $O(1)$

Syntax :-

Variable - name - dictionary [key] = value

6_Data Structure in Python > 09_Dictionaries > 01_Creating Dictionaries.py > ...

```
1  # Dictionary
2  thisdict = {
3      "brand": "Ford",
4      "model": "Mustang",
5      "year": 1964
6  }
7  print(thisdict)
8
9  # Dictionary Items
10 print(thisdict["brand"])
11
12 # Duplicates Not Allowed
13 thisdict = {
14     "brand": "Ford",
15     "model": "Mustang",
16     "year": 1964,
17     "year": 2020
18 }
19 print(thisdict)
20
21 # Dictionary Length
22 print(len(thisdict))
23
24 # Dictionary Items - Data Types
25 thisdict = {
26     "brand": "Ford",
27     "electric": False,
28     "year": 1964,
29     "colors": ["red", "white", "blue"]
30 }
```



Ln 22, Col 21

Spaces: 2

UTF-8

CRLF

{ } Python

Co

02_Access Dictionary Items.py X

6_Data Structure in Python > 09_Dictionaries > 02_Access Dictionary Items.py > ...

```
1  thisdict = {
2      "brand": "Ford",
3      "model": "Mustang",
4      "year": 1964
5  }
6  x = thisdict["model"]
7  print(x)
8
9  # There is also a method called get()
10 # that will give you the same result:
11 x = thisdict.get("model")
12 print(x)
13
14 # Get Keys
15 x = thisdict.keys()
16 print(x)
17
18 # Add a new item to the original dictionary,
19 thisdict["color"] = "white"
20 print(x)
21
22 # Get Values
23 x = thisdict.values()
24 print(x)
25
26 # Make a change in the original dictionary,
27 thisdict["year"] = 2025
28 print(x)
29
30 # Add a new item to the original dictionary.
```


03_Change Dictionary Items.py ✕

6_Data Structure in Python > 09_Dictionaries > 03_Change Dictionary Items.py > ...

```
1  thisdict = {
2      "brand": "Ford",
3      "model": "Mustang",
4      "year": 1964
5  }
6  thisdict["year"] = 2018
7  print(thisdict)
8
9  # Update Dictionary
10 thisdict.update({"year": 2025})
11 print(thisdict)
```

6_Data Structure in Python > 09_Dictionaries > 04_Add Dictionary Items.py > ...

```
1  thisdict = {
2      "brand": "Ford",
3      "model": "Mustang",
4      "year": 1964
5  }
6  thisdict["color"] = "red"
7  print(thisdict)
8
9  # Update Dictionary
10 thisdict.update({"color": "white"})
11 print(thisdict)
```



```
1  # The pop()
2
3  thisdict = {
4      "brand": "Ford",
5      "model": "Mustang",
6      "year": 1964
7  }
8  thisdict.pop("model")
9  print(thisdict)
10
11 # The popitem() method
12 thisdict.popitem()
13 print(thisdict)
14
15 # The del keyword removes the item with the specified key name:
16 del thisdict["brand"]
17 print(thisdict)
18
19 # The del keyword can also delete the dictionary completely:
20 del thisdict
21 print(thisdict)
22
23 # The clear() method empties the dictionary:
24 thisdict.clear()
25 print(thisdict)
```


06_ Loop Dictionaries.py X

6_Data Structure in Python > 09_Dictionaries > 06_ Loop Dictionaries.py > ...

```
1  # Print all key names in the dictionary, one by one:
2
3  thisdict = {
4      "brand": "Ford",
5      "model": "Mustang",
6      "year": 1964
7  }
8  for x in thisdict:
9      print(x)
10     # Print all values in the dictionary, one by one:
11     print(thisdict[x])
12
13     # You can also use the values() method to return values of a dictionary
14     for x in thisdict.values():
15         print(x)
16
17     # You can use the keys() method to return the keys of a dictionary:
18     for x in thisdict.keys():
19         print(x)
20
21     # Loop through both keys and values, by using the items() method:
22     for x, y in thisdict.items():
23         print(x, y)
24
25     # Copy Dictionaries
26     # with the copy() method:
27     mydict = thisdict.copy()
28     print(mydict)
29
30     # with the dict() function:
```


data structure in Python > 09_Dictionaries > 07_Nested Dictionaries.py > ...

```
1 myfamily = {  
2     "child1" : {  
3         "name": "rashti",  
4         "year": 2028  
5     },  
6     "child2" : {  
7         "name": "sruja",  
8         "year": 2029  
9     },  
10    "child3" : {  
11        "name": "rashi",  
12        "year": 2030  
13    }  
14 }  
15 print(myfamily)
```

OR

```
1 child1 = {  
2     "name" : "Emil",  
3     "year" : 2004  
4 }  
5 child2 = {  
6     "name" : "Tobias",  
7     "year" : 2007  
8 }  
9 child3 = {  
10    "name" : "Linus",  
11    "year" : 2011  
12 }
```